IPCON® PP GF

Glass Fiber Reinforced Polypropylene Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.03 mm
Net Filament Weight	0.75 kg ± 0.5%
Filament Length	300 ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C) or PC (max heat resistance: 130 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm
Recommendation for body protection during use	This material contains carbon fiber, which may cause skin irritation. So it is recommended to wear gloves throughout the operation.

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.04 g/cm³
Shrinkage Rate*	Specially-defined	0.25% - 0.57%
Surface Hardness	Shore hardness	70D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	< 0.06%
Melt Index	230 °C, 2.16 kg	11.9 g/10 min
Melting Temperature	DSC, 10 °C/min	155 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	Not annealed: 102 °C Annealed: 113 °C*
Heat Deflection Temperature	ISO 75, 0.45 MPa	Not annealed: 98 °C Annealed: 104 °C*
Heat Deflection Temperature	ISO 75, 1.80 MPa	Not annealed: 84°C Annealed: 89 °C*
Relative Thermal Index (RTI) Max Continuous Service Temperature	Actual working conditions simulation test	70 - 85 °C (Related to load, size, shape, wall thickness, etc., and the value after annealing is higher than that before annealing in general)

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	31 MPa
Tensile Strength Z	ISO 527, GB/T 1040	17 MPa
Young's Modulus XY	ISO 527, GB/T 1040	1268 MPa

Young's Modulus Z	ISO 527, GB/T 1040	635 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	26.5%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	25.4%
Flexural Strength XY	ISO 178, GB/T 9341	53 MPa (No Break)
Flexural Strength Z	ISO 178, GB/T 9341	31 MPa (No Break)
Flexural Modulus XY	ISO 178, GB/T 9341	1464 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	664 MPa
Impact Strength XY	ISO 179, GB/T 1043	46.5 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	13.4 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 240 °C, bed temperature: 60 °C, printing speed: XY—147 mm/s and Z—70 mm/s) and IPCON PP GF (Grey) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The XY specimens' annealing parameter is: 60 °C, 4 h → 80 °C, 8 h. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Ultra-low density, extremely lightweight; hydrophobic, oil and grease-resistant, and extremely resistant to chemical corrosion; high toughness and layer adhesion strength, fatigue and flex-resistant; wide service temperature range (approximately -20 °C - 100 °C); carbon fiber reinforced, smooth and delicate surface; less warping, easier to print, compatible with mainstream printers and multi-color feeding systems such as AMS and CFS.
2. It can be used to print vehicle brackets and accessories for toy racing cars, drones, and robots that require lightweight, chemical reagents handling tools, electroplating fixtures, and cleaning benches that require resistance to acids, alkalis, solvents, and greases, and buckles, hinges, and movable joints that require fatigue and flex resistance, electronic component enclosures that required insulation.

• Recommended Printing Settings

Drying Settings Before Printing	70 - 100 °C, 6 - 8 h (Blast drying oven)
Printing Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself, PLA or support filament for PP
Compatible Printer Type	Open-frame, enclosed-frame
AMS or CFS	Compatible
Compatible Nozzle Material	Hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.4 / 0.6 / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plates
Plate Surface Preparation	Glue for PP must be sprayed or film for PP must be pasted
Nozzle Temperature	220 - 260 °C

Bed Temperature	55 - 75 °C
Fan Speed	10% - 70%
Max Print Speed	250 mm/s (P2S, 235 °C) (For the thin and high models and the support structure, the recommended speed is lower than 150 mm/s)

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